

Project:

Superstore Sales

Analysis & Forecasting

by Power BI

Prepared by: Hager Mohamed Siddiq

Email: ha9armohamed@gmail.com

LinkedIn: www.linkedin.com/in/hager-mohamed-siddiq

Project Objective:

To analyze sales data and identify trends and patterns in sales performance, customer behavior, product categories, shipping methods, and geographic markets to support data-driven business decisions.

Dateset Overview:

Attribute	Value
Sales Period	2015 - 2018 (4 Years)
Categories	Furniture, Office Supplies, Technology
Segments	Home Office, Corporate, Consumer
Ship Modes	Standard Class, Second Class, First Class, Same Day
Country	United State

Key KPIs:

KPI	Value
Total Sales	2,259,610
Total Orders	4,922
Total Customers	793
Total Products	1,861
Total States	49
Total Cities	529

Analysis:

- **Sales Analysis**

- **2018 was the highest-performing sales year**, accounting for **31.9% of total sales** over the four-year period. The year generated total sales of **721,351**.
- In addition, 2018 recorded product shipments to **47 out of 49 states**, indicating the broadest geographic reach among the four years analyzed.
- From a geographic perspective, the **West region** generated the highest share of total sales, accounting for **31.4%** with **710,220 of Sales**.
- Among product categories, **Technology** recorded the highest sales contribution, representing **36.5% of total sales** with **825,528 of Sales**.
- By customer segment, **Consumer** was the highest-performing segment, accounting for **50.8% of total sales** with **1,147,769 of Sales**.
- In terms of shipping methods, **Standard Class** generated the highest share of sales, representing **59.3% of total sales** with **1,339,430 of Sales**.

- **Orders Analysis**

- **2018 recorded the highest number of orders**, with **1,661 orders**, representing **33.7% of total orders** during the analyzed period.
- The **West region** recorded the highest share of orders, accounting for **32.2% of total orders** with **1587 Orders**.
- Among the product categories, **Office Supplies** generated the highest number of orders, representing **74.7% of total orders** with **3676 orders**.
- By customer segment, **Consumer** recorded the highest share of orders, accounting for **51.5% of total orders** with **2537 orders**.

- **Product Analysis**

- The **10 best-selling products are not the same as the 10 most-ordered products**, as shown on the second page of the Power BI dashboard.
- The **best-selling product** generated **61,599.824 in sales** from only **5 orders**, while the **most-ordered product** recorded **47 orders** but generated only **1,675,652 in sales**.

- Overall, the **10 best-selling products** generated total sales of **244,620.204**, accounting for **10.82% of total sales**, across **64 orders**.
- In comparison, the **10 most-ordered products** recorded **273 orders**, representing **5.5% of total orders**, with total sales of **16,003.681**.
- **Customer Analysis**
- The **10 highest-selling customers** are also not the same as the **10 most-ordered customers**.
- The **10 highest-selling customers** generated total sales of **153,811.169**, accounting for **6.80% of total sales** generated over the four-year period, across **73 orders**.
- In comparison, the **10 most-ordered customers** generated total sales of **66,321.8148**, with a total of **115 orders**. These sales represent **2.93% of total sales**, while the orders account for **2.33% of total orders**.
- **Four-Year Sales Analysis**

The sales performance over the four-year period was as follows:

Year	Total Sales
2015	479,331
2016	459,436
2017	599,492
2018	721,351

- **2015 recorded the lowest sales** among the four years, while **2018 recorded the highest sales**.
- The **first quarter (Q1)** of each year consistently recorded the **lowest sales** throughout the four-year period.
- The **fourth quarter (Q4)** consistently recorded the **highest sales** in each year.
- Within Q4, sales consistently increased in **November**, followed by **December**. This pattern highlights the **impact of seasonality on sales growth**.
- During the **first two years**, sales in **Q2** were nearly identical, with only a slight difference between the two years.

- During the **last two years**, Q2 sales increased compared with the first two years; however, the sales figures for **2017 and 2018 remained relatively close to each other**.
- Sales were generally **relatively consistent across most months**. However, over the four-year period, noticeable increases occurred during **March, September, and November**.
- **February recorded the lowest sales** among the months, followed by **January**.
- **October sales remained stable during the first two years**.
- These trends are clearly illustrated in the **Line Chart** on the dashboard and the **Matrix Table** on the project's third page.

2019 Sales Forecast

Based on the sales patterns observed over the previous four years, sales are expected to increase during 2019, particularly in the periods that consistently showed higher sales activity.

The expected increases are particularly likely to occur in **March**, during the **first quarter (Q1)**; **September**, during the **third quarter (Q3)**; and **November and December**, during the **fourth quarter (Q4)**.

The year-over-year sales growth also indicates a positive overall trend. Sales in **2016 increased by 4.15% compared with 2015**, while sales in **2017 increased by 30.48% compared with 2016**, and sales in **2018 increased by 20.33% compared with 2017**.

Based on these historical sales patterns and the sales performance observed over the four-year period, **2019 is expected to achieve sales growth of at least 25% compared with 2018**.

This forecast is based on the analysis of the four previous years, including historical sales trends, recurring seasonal patterns, and the overall sales volume achieved during the analyzed period.

Conclusion

The analysis of the four-year sales dataset revealed a consistent overall growth trend, with **2018 recording the highest sales performance** and the highest number of orders during the analyzed period.

The analysis also showed that sales performance varied across regions, product categories, customer segments, and shipping methods. The **West region, Technology category, Consumer segment, and Standard Class shipping method** recorded the highest shares of sales.

The analysis of products and customers revealed an important distinction between **sales value and order volume**. The products and customers generating the highest sales were not necessarily the same as those generating the highest number of orders.

The four-year analysis also identified recurring seasonal patterns, particularly higher sales activity in **March, September, November, and December**, with the fourth quarter consistently recording the strongest sales performance.

Based on these historical trends and recurring patterns, **sales are expected to continue growing in 2019**, with higher sales activity particularly anticipated during the previously identified high-performing periods.

Explore the complete project, including the Power BI dashboard and analysis, on GitHub.

My GitHub Account: [Ha9ar \(Hager Mohamed Siddiq\)](#)